

# Summary - note\_5.pdf

Artificial Intelligence (AI) is a field dedicated to creating machines that mimic human cognitive functions such as learning, problem-solving, and decision-making, enabling them to perceive, reason, and act intelligently within complex environments.

Key areas of AI include:

- \* **Machine Learning (ML):** Algorithms learn patterns from data for predictions without explicit programming.
- \* **Deep Learning (DL):** A specialized ML area using multi-layered artificial neural networks for complex data processing.
- \* **Natural Language Processing (NLP):** Allows machines to understand, interpret, and generate human language.
- \* **Computer Vision (CV):** Enables machines to interpret visual information.
- \* **Robotics:** Integrates AI for intelligent control and decision-making in physical robots.

AI is widely applied across sectors like healthcare, finance, manufacturing, and education. It is a transformative and rapidly evolving field promising significant benefits, while also necessitating continuous ethical considerations for responsible development.